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Analysis of Violent Crime Across U.S. States: Urban Population and Regional Differences

Introduction

Violent crime is a major societal concern, and understanding its distribution can help inform policies, law enforcement strategies, and resource allocation. In this study, I investigate whether there is more violent crime in the Northeast than in the Midwest, and whether this pattern is associated with differences in urban population across states.

The hypothesis for this analysis is that states in the Northeast will have higher total violent crime rates than states in the Midwest, largely due to the generally higher urban population in the Northeast. Understanding these patterns is important because it highlights whether regional differences are driven by geography alone or by demographic characteristics such as urban density.

Methods

The analysis used the *USArrests* dataset, which contains violent crime data and urban population percentages for all 50 U.S. states in 1973. The primary variables were murder rate, assault rate, rape rate, and urban population. To create a single measure of violent crime, the three crime types were summed to produce a total crime rate for each state. States were categorized into two regions for comparison: the Northeast, including states such as New York, Massachusetts, and Pennsylvania, and the Midwest, including Ohio, Illinois, and Michigan.

The analytical approach included several steps. First, descriptive statistics, including mean, median, mode, range, and standard deviation, were calculated for each crime type, urban population, and total crime rate. Second, visualizations were created, including a histogram of total crime, boxplots and bar plots comparing regions, and a scatterplot to examine the correlation between urban population and total crime. Pearson correlation was used to quantify the strength and significance of the relationship between urban population and total crime. Finally, a linear regression model was fit to predict the total crime rate from region and urban population, allowing for a formal test of the hypothesis while controlling for urbanization.

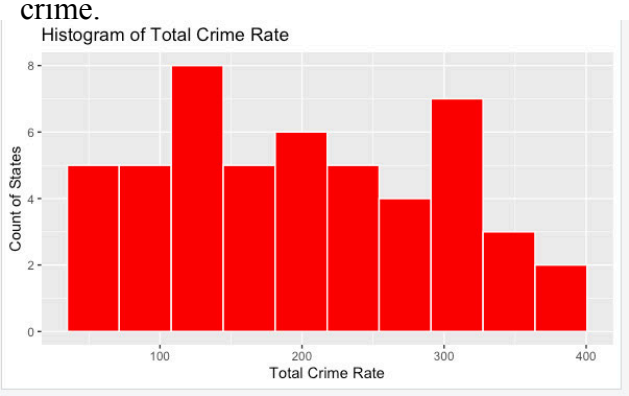
Results

Descriptive Statistics

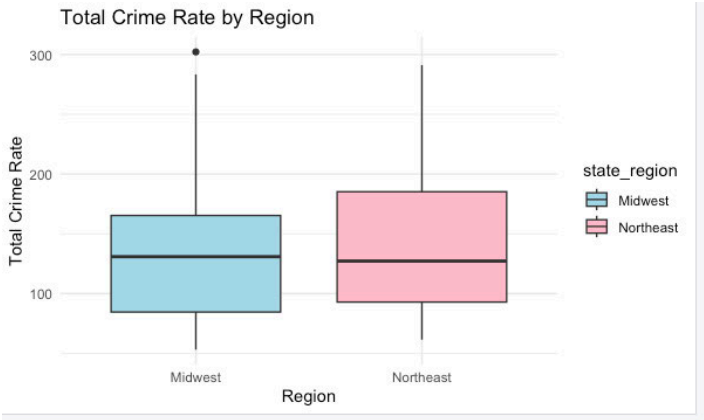
Descriptive statistics revealed notable variation across states. The mean murder rate was 7.79 per 100,000 residents, with a median of 7.25, and several modes clustered around lower values. Assault rates were higher and more variable, with a mean of 170.76 and a median of 159. Rape rates averaged 21.23, with a median of 20.1, and urban population ranged from 32% to 91%, averaging 65.54%. Total crime rates ranged from approximately 53 to 400, with a mean of 199.78 and median of 185.2. These descriptive statistics indicate that assault contributes most to the variability in total crime, while urban population varies moderately**.**

Visualizations

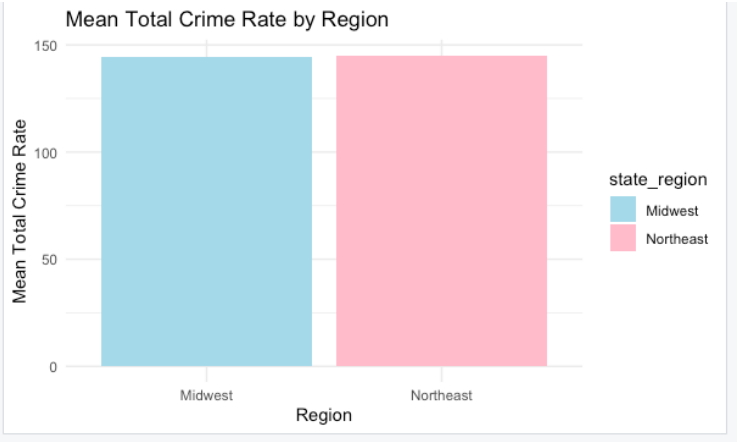
Histogram of total crime rate: The histogram illustrates the distribution of total crime rates across all states, showing a disorganized pattern with several states experiencing unusually high crime.



Boxplot by region: Boxplots comparing the Northeast and Midwest reveal that the Northeast has a slightly higher median total crime rate and a wider range of values, while the Midwest is more consistent.



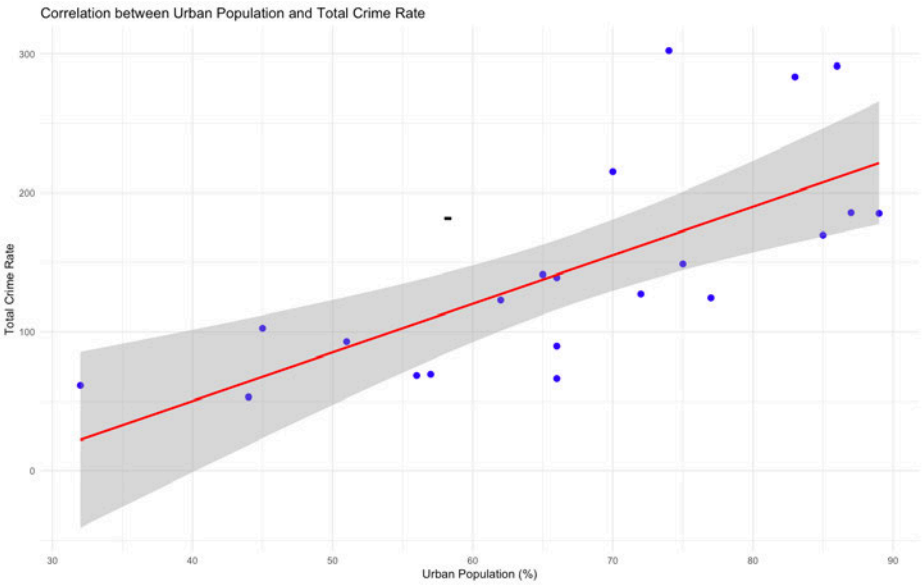
Bar plot of mean total crime rate by region: The mean total crime in the Northeast is slightly higher than in the Midwest.



Correlation Analysis

Pearson's correlation between urban population and total crime rate is 0.712 ($p < 0.001$). This indicates a strong positive association: states with higher urban populations tend to have higher total crime.

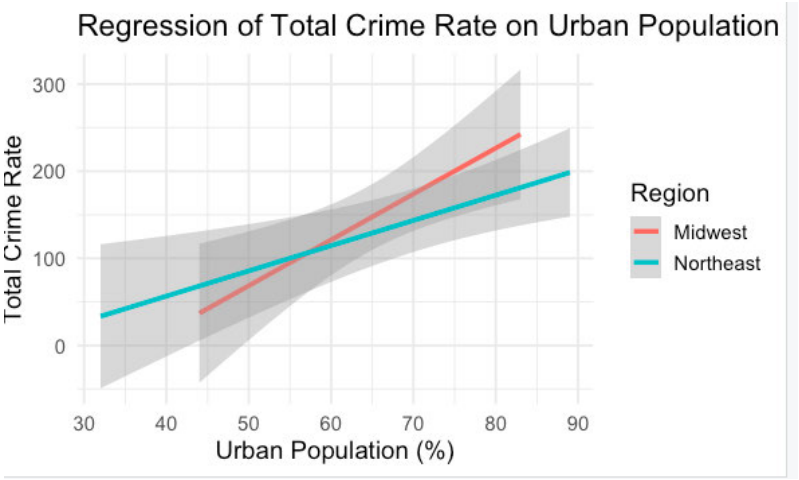
A scatterplot with a regression line illustrates this trend, confirming that urban population is strongly linked to total crime rates across states.



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Regression Analysis

Linear regression further clarified these patterns. When total crime was regressed on both region and urban population, urban population emerged as a highly significant predictor. Specifically, each one percent increase in urban population was associated with a 3.64 increase in total crime. Region itself was not a significant predictor; after controlling for urban population, the Northeast did not exhibit significantly higher crime than the Midwest. The R^2 of 0.527 indicates that more than half of the variation in total crime is explained by these two factors. Regression plots show parallel lines for the Northeast and Midwest, illustrating that urban population affects crime similarly in both regions.



Discussion

The results provide insight into the drivers of violent crime across U.S. states. Urban population is a strong predictor of total crime, confirming the expectation that more urbanized states tend to experience higher crime rates. While descriptive statistics initially suggested that the Northeast had slightly higher total crime than the Midwest, regression analysis indicates that this difference is not statistically significant once urban population is accounted for. Therefore, the initial hypothesis that the Northeast would have higher crime is only partially supported.

Assault rates were found to be the most variable crime type, largely driving differences in total crime across states. The findings emphasize that demographic factors, rather than region alone,

are the main determinants of crime variation. For policymakers and law enforcement, this suggests that focusing resources on urban areas could be more effective than targeting regions based solely on geography.

In conclusion, the analysis demonstrates that urban population significantly influences total violent crime, while regional differences between the Northeast and Midwest are largely explained by urbanization. As the researcher, I initially expected higher crime in the Northeast, but the evidence indicates that urban population is the key factor, and the Northeast is not inherently more violent than the Midwest. These findings highlight the importance of considering demographic factors when evaluating regional differences in crime.